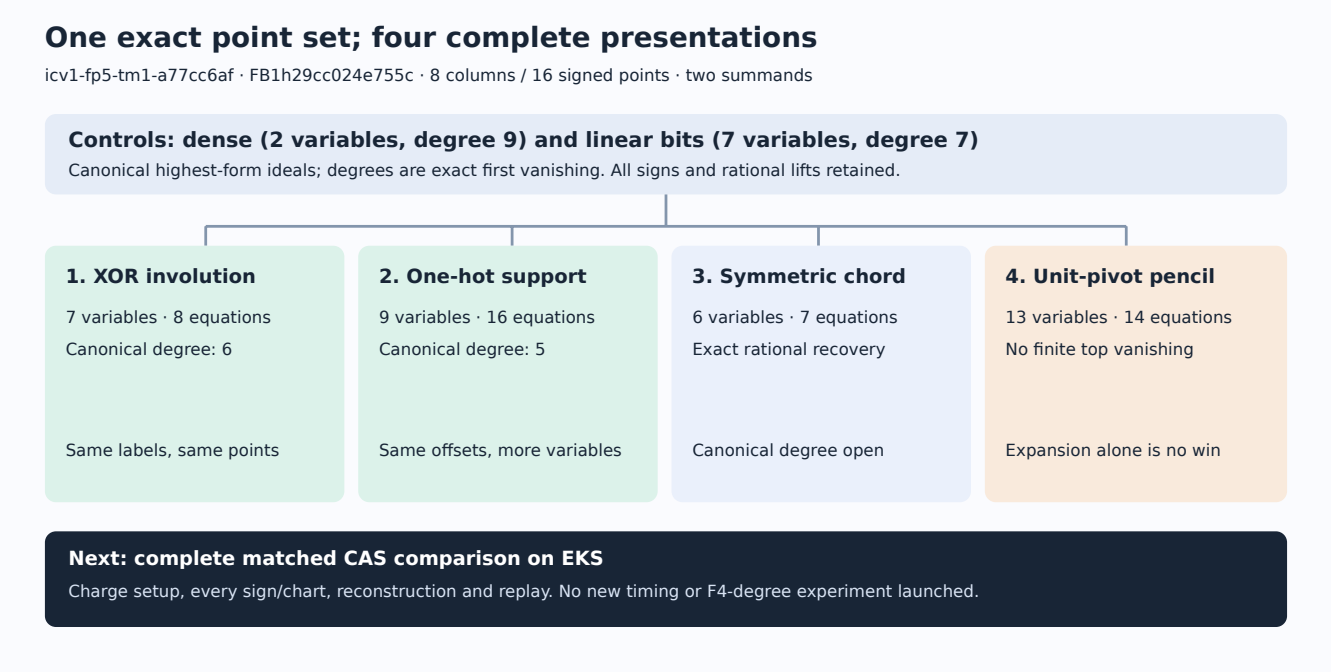
Four complete selector presentations

Same factor base; two exact canonical degree reductions

2026-11-01 · formal algebra and development correctness checks

The existing F17 two-summand factor-base system has two better canonical presentations: XOR recoding gives exact degree 6 with seven variables, and one-hot support gives exact degree 5 with nine variables. The original labels give degree 7; the same-base dense presentation gives degree 9. All selector variables and support constraints are counted.

Identity. icv1-fp5-tm1-a77cc6af, curve y^2=x^3+2x+2 over F17; FB1h29cc024e755c. Eight columns, sixteen signed points. Reuse the original issuer and FB1 dump. This is a new presentation of the same points, with no newly registered curve.



Meaning of degree. The exact first degree at which the entire homogeneous component lies in the ideal generated by the actual equations' canonical highest forms. It is not an observed affine F4 solving degree, runtime, yield or rank measurement.

Presentation	Vars	Degree	Ambient count
Dense control	2	9	55
Linear-bit control	7	7	3432
XOR involution	7	6	1716
One-hot support	9	5	2002

Ambient counts are derived choose(N+D,N) values, checked in Envelope.lean. They are not measured F4 matrix widths. The lower-degree one-hot arm has a larger ambient count than XOR.

1. Algebra and exact semantics

1. XOR.

Boolean replacement $(u,v) \to (u+v-2uv,v)$ is an involution. The offset becomes $2u+12v-4uv$ and retains exactly $\{0,2,10,12\}$. The two graph equations and four Boolean equations remain in the seven-variable system.

After invertible row normalization, the top ideal is generated by m^2, xz, $x^2-amx-uv$, $z^2-bmz-wt$, and the four bit squares. Every degree-6 monomial vanishes. The functional with support $xmuvw$, x^2uvw, x^3mw, x^4w and weights $1,a,1,2a$ annihilates the ideal but takes value 1 on the degree-5 witness x^3mw. Both bounds are checked over a field. RecodedCanonical.lean connects this ideal to the actual generator set; FourTrackFixture.lean instantiates the F17 system and complete finite-target point semantics.

2. One-hot.

Three bits per slot satisfy Boolean equations and all same-slot pair products. Exactly 000, 100, 010, 001 remain; weights 2, 10, 12 give the original offsets. At most one bit per slot survives in the top quotient. Together with degree-3 core vanishing this gives the degree-5 upper bound. Retaining one bit per slot maps to the old two-bit quotient; its degree-4 witness remains nonzero when characteristic is not 2. SupportOneHot.lean checks both bounds and the actual highest-form bridge.

3. Symmetric chord.

Set $\sigma=m^2-r$, $\pi=r^2+rm^2+A+2ms$, and $D=\sigma^2-4\pi$. Recover $x=(\sigma+d)/2$, $z=(\sigma-d)/2$. Two incidence equations are linear in d, with $d^2=D$. SymmetricChord.lean proves equivalence and covers three disjoint cases: first pivot nonzero; first zero and second nonzero; both zero with both constants zero and a square root in the base field. The six-variable native system retains d; no saturated chart is counted as a complete search. Its canonical degree remains open.

4. Determinant.

A 3×3 pencil with bottom-right identity block has determinant $y-(ax^2+bx+c+l_0+l_1l_2)$. Its normalized kernel is $[1,-ax,-l_2]$. This chart is complete. Choosing $l_0=2u+12v$, $l_1=-4u$, $l_2=v$ returns XOR membership. Determinantal.lean checks the equivalence.

The expanded 13-variable system loses both ordinate variables from its highest forms. Evaluation sending one ordinate to 1 and all other variables to 0 annihilates the whole top ideal but every power of that ordinate maps to 1. DeterminantalTop.lean proves no finite homogeneous first vanishing, including arbitrary affine lower terms. This does not assert that its affine solution set is infinite. Schur elimination returns the seven-variable XOR system; it is a separately priced reduction.

2. Checks, sources and next experiment

The native Rust implementation emits all six model families. The known-answer target (13,10) yields 21 complete inputs, with msolve syntax, sparse F17 polynomial coefficients, exact ordered group references and hashes. Uncompressed metadata SHA256: 154c40dcea500bfe76a7eb9d8bb411e42e4106407d0ad6bf909a7b8528d7c87c

Seven release correctness tests check arithmetic, unchanged labels/points, complete scalar reference against group replay on all 18 finite targets, symmetric and determinantal evaluations, dense/group equivalence, every F17 linear-square pivot case, and one-hot support with group replay. Correctness-test harness time is not a benchmark. All performance/rank counters in the export are null.

verify.sh checks source hashes, the Lake project, warnings-as-errors, the unchanged whole transitive axiom auditor, native clippy/formatting/tests and byte-identical fixed-fixture export. Lean is pinned to 4.19.0 and Mathlib to c44e0c8ee63ca166450922a373c7409c5d26b00b. The source-linked Markdown report, proof map and verification/ receipts are the detailed record. Failed development logs and source snapshots are retained separately from accepted sources; no sorry, custom axiom or native trust is accepted.

The broad root-library attempt under installed Rust 1.94.1 reaches the two existing E0658 atomic_try_update errors in src/cryptanalysis/ecbench/pencil.rs and src/cryptanalysis/ecbench/diagnostic_runtime.rs. No unrelated source workaround or root-library pass is claimed. Use a compatible root toolchain and current-head CI for that scope.

Prospective comparison. PROTOCOL.md freezes a complete finite-target panel, all four signs, exact decoding/replay, A/A controls and all setup/branch/reconstruction work. It requires matched msolve F4 and the fastest licensed reference when available, with matrix dimensions, degrees, arithmetic, RAM and disk counters. No new CAS/time/yield experiment has launched. The current EKS route lacks an arbitrary standalone CAS adapter and ready identity; host and budget admission remain to be frozen. Original 17-term P-256 gates remain separate obligations.

Literature checked before implementation.

- Petit–Kosters–Messeng (2016), Algebraic approaches for the ECDLP over prime fields. Factor-base constructions and experimental-degree scope; composed/Dickson methods are prior art.
- Faugère–Safey El Din–Spaenlehauer (2011), Gröbner bases of bihomogeneous systems. Genericity and zero-dimensional hypotheses require verification; bilinear syntax alone does not supply them.
- Repository 24-design register, previous Boolean-offset proof, and parallel_20261031/RESULTS.md: earlier local degree reductions did not reduce complete branch-summed work against dense.

No external novelty or solve-time gain is asserted. The completed outcomes are two exact degree reductions, complete rational symmetric reconstruction, and the determinantal expansion obstruction. The next technical step is the complete matched solver comparison under the frozen protocol.